ANLY 656 Midterm Report

Predicting Sleep Deficit Among Software Engineers

Data: sleep_rating.csv (8,500 employees, 16 attributes)
Target: sleep_deficit — nominal classes 0-none, mild, moderate, severe
Objective: Predict sleep-deficit level and identify attribute effects that help address its causes.
Programs: log_reg.py, tree.py, forest.py (with matching *_output.txt files)

1. Problem and Approach

Sleep deficit among software engineers affects health and productivity. This analysis builds three classification models studied in ANLY 656 — multinomial logistic regression, a decision tree, and a random forest — using the same cleaned and encoded data.

Data preparation (shared across models)

- Data map drafted with Data_Map_Tool.py and reviewed.
- morning_alarm_snoozes treated as Interval (ordered count).
- Nominal one-hot encoding for gender, occupation, favorite bedtime app, and clock type.
- Target kept as class labels (DT.String).
- ReplaceImputeEncode (RIE): 0 missing values and 0 outliers within map bounds.
- Encoded design matrix: 28 predictors (forest later reduced features for tuning).

Class imbalance: moderate 52.5% · mild 23.6% · 0-none 16.3% · severe 7.6%. All hold-out and CV splits used stratification. The rare **severe** class is hardest to recover. Validation follows the course checklist using misclassification (MISC) and the ratio MISC_validation / MISC_training; ratios above **1.2** are flagged as possible overfitting.

2. Model Results

2.1 Logistic Regression (log_reg.py)

RIE used interval_scale="std". Multinomial logistic was compared with One-vs-Rest; **Multinomial** was selected (97.89% vs 87.88% full-data accuracy) and recovered severe cases much better on full data (≈86.6% vs ≈62.8% sensitivity).

Metric	Training (70%)	Validation (30%)
Accuracy	98.02%	97.53%
MISC	2.0%	2.5%
Overfit ratio	—	1.25

10-fold CV test MISC ≈ 2.5% (mean overfit ratio ≈ 1.20). Validation sensitivity: 0-none 99.3%, mild 99.0%, moderate 98.4%, **severe 83.5%** (most often mistaken for moderate). Coefficient signals: lower total_sleep_hours, higher sleep_latency_min, and more morning_alarm_snoozes associate with worse deficit classes.

2.2 Decision Tree (tree.py)

RIE used no interval standardization. A kitchen-sink tree overfit (perfect train fit; 3.8% val MISC). Grid search selected **gini, max_depth=7, min_samples_leaf=20, min_samples_split=20** (37 leaves).

Metric	Training (70%)	Validation (30%)

Accuracy	97.70%	97.14%
MISC	2.3%	2.9%
Overfit ratio	—	1.24

Best CV: 4-fold, val MISC ≈ 2.71% (ratio ≈ 1.21). Importance: total_sleep_hours (~87%), morning_alarm_snoozes (~10%), sleep_latency_min (~2%), with smaller roles for deep-sleep %, caffeine, brightness, and blue-light filter.

2.3 Random Forest (forest.py)

Per exam guidance, feature selection was used: features accounting for 99% of decision-tree importance (**9 of 28** predictors). The case count was not reduced. Grid search selected **entropy, n_estimators=50, max_depth=7, max_features=6, min_samples_leaf=39, min_samples_split=78**.

Metric	Training (70%)	Validation (30%)
Accuracy	97.71%	97.80%
MISC	2.3%	2.2%
Overfit ratio	—	0.96

Best CV: 9-fold, val MISC ≈ 2.51% (ratio ≈ 1.19). Importance: total_sleep_hours (~82%), morning_alarm_snoozes (~14%), sleep_latency_min (~3%), plus smaller effects from bedtime phone minutes and deep-sleep %.

3. Comparison

Model	Val Acc	Val MISC	Overfit	Severe notes	Interpretability
Multinomial logistic	97.53%	2.5%	1.25	Val sens. 83.5%	Signed coefficients
Decision tree	97.14%	2.9%	1.24	Severe MISC ≈20%	Rules / tree plot
Random forest	97.80%	2.2%	0.96	Severe MISC ≈20%	Importance ranks

All three models agree that sleep quantity and bedtime/wake behaviors drive deficit class: **total sleep hours**, **morning alarm snoozes**, and **sleep latency**.

4. Strengths and Shortcomings

Logistic regression — Strengths: Clear signed coefficients for interventions; multinomial beats OvR; strong overall accuracy; useful per-class sensitivity. **Shortcomings:** Hold-out overfit ratio slightly above 1.2; one-hot/collinear nominal blocks complicate coefficient interpretation; assumes linear log-odds.

Decision tree — Strengths: Transparent split rules and plot; importance aligns with other models; kitchen-sink vs tuned contrast shows value of regularization. **Shortcomings:** Highest validation MISC; overfit ratio still above 1.2; single tree less stable than an ensemble; weak on rare severe class.

Random forest — Strengths: Best hold-out accuracy and lowest MISC; overfit ratio under 1.2; feature selection shortened the model without dropping cases; stable CV. **Shortcomings:** Less directly interpretable than logistic coefficients or a single tree; heavier grid search; still struggles with severe vs moderate.

5. Preferred Model

Preferred model: Random Forest. It has the best validated accuracy/MISC on the 70/30 hold-out (97.8% / 2.2%), shows no overfitting by the course ratio rule (0.96 vs 1.25 logistic and 1.24 tree), and yields a short actionable feature list after selection that matches the cause story from the other models. Exam guidance explicitly allows forest feature

selection, which was used without shrinking the sample.

Logistic remains valuable for signed effect interpretation in a follow-up briefing. The tree is useful for communicating simple rules. For deployment of a single predictive model on these data, the forest is the best balance of accuracy and generalization.

6. Suggestions for Expanding This Work

- Cost-sensitive or balanced learning for the rare severe class (class weights, threshold tuning, or focused sampling).
- Ordered/ordinal models treating deficit severity as ordered rather than unordered nominal.
- Intervention simulation — estimate how changing sleep hours, snoozes, or latency shifts predicted class probabilities.
- External validation on a later employee cohort or another company site.
- Policy packaging — combine forest scores with logistic coefficients into an employee wellness checklist.

7. Deliverables Checklist

File	Role
log_reg.py / log_reg_output.txt	Logistic solution + output
tree.py / tree_output.txt	Decision tree solution + output
forest.py / forest_output.txt	Random forest solution + output
midterm_report.pdf	This summary report

Supporting plots (confusion matrices, tree diagram, forest importance) are saved alongside the programs in the Midterm project folder.